

Course No.	Title of the course	Course Structure	Pre-requisite
EPMTC301	Mathematics for Machine Learning	3L-1T-0P	None

COURSE OUTCOMES:

1. To understand the vector space and linear transformation.
2. To know the concept of inner product spaces and norms.
3. To know concept of matrix decomposition.
4. To know concept of vector calculus.
5. To know concept of optimization techniques.

COURSE CONTENT:

UNIT-I

Linear Algebra: Vector space, Subspaces, linearly dependent and independent vectors, Basis and dimension of a vector space, Linear Transformation with examples, Rank-Nullity theorem, Matrix representation of Linear Transformation.

UNIT-II

Matrix Theory: Inner Product Space, Norms, Orthogonal Component and Projection, Gram-Schmidt process, Properties of Eigen value and Eigen vector, Orthogonal Matrix, Symmetric & Skew Symmetric Matrix, Quadratic form, Positive Definite Matrix, Positive Semi Definite Matrix.

UNIT-III

Matrix Decomposition: Singular Value Decomposition, Pseudo-Inverse, Low Rank Approximation, Polar decomposition, Right Polar Decomposition, Left Polar Decomposition.

UNIT-IV

Vector Calculus: Directional derivatives, Hessian, Convex set & function and its properties; Scalar and vector functions, Gradient, Divergence and Curl, Directional Derivatives, Line Integrals, Surface Integrals, Volume Integrals, Green's Theorem, Stoke's Theorem and Gauss Divergence Theorem (without proof).

UNIT-V

Optimization: Optimality conditions for unconstrained optimization, Constrained optimization problem, KKT optimality conditions, Convex and Quadratic Programming Problem, Classification problems, Support vector machine, Steepest descent method.

SUGGESTED BOOKS:

1. M. P. Deisenroth, A. A. Faisal & C. S. Ong, "Mathematics for machine learning," Cambridge University Press, 2020
2. M. Pradhan and U. D. Kumar, "Machine Learning using Python", 2019.

